

Panagiotis Lymperopoulos

PhD Candidate in Computer Science

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Education

Tufts University

PhD Candidate in Computer Science

MS in Data Science

University of California, Los Angeles (UCLA)

Bachelor of Science in Computational and Systems Biology

Top AI Publication Venues: AISTATS, AAMAS, NeurIPS Workshop, EMNLP Workshop, ICLR Workshop, NeSy, TMLR, AIJ

Skills: Python, PyTorch, TensorFlow, Reinforcement Learning, Deep Learning, Graph Neural Networks, **RAG**, LLMs, LLM Agents, Agile Development.

Work Experience

Research Intern, Skyfall, New York City, US

June 2025 - August 2025

- Developed World Models for LLM-based planning agents by integrating foundation models' implicit world knowledge with external and domain-specific data.
- Combined LLM-designed probabilistic graphical models with code generation to automatically design and optimize sparse world models for planning with MCTS.

LLM Engineer, Agora Labs, Rome, Italy (Remote)

June 2024 - May 2025

- Led team of developers in development and deployment of customer-facing text-to-SQL and automated data preprocessing application.
- Designed and implemented information extraction pipeline for medical data, integrating traditional NLP workflows (NER, entity-linking) with LLMs.
- Focused on ensuring data privacy and security by deploying and fine-tuning local, open-source LLMs.

Data Science Intern, Optechain, Athens, Greece

June 2021 - August 2021

- Implemented ML pipeline for Data Cleaning, Training and Deployment of time-series prediction and computer-vision models on AzureML
- Developed infrastructure on the entire stack for querying deployed models, submitting results to database and serving them to end-users.

Research Highlights

Act-to-Ground: Neurosymbolic Grounding for Planners

September 2023 - Present

- Developed Act-to-Ground (A2G), a framework for training neural grounding models for symbolic planners using weak supervision from environment interaction or demonstrations. Motivated by a need to separate perception from action to ensure safe and interpretable decision making.
- Proposed a novel MCMC sampler and a score-matching objective to improve the stability and performance of neurosymbolic grounding models.
- Presented the framework and its successful evaluation in planning domains at the 19th Conference on Neurosymbolic Learning and Reasoning (NeSy 2025).

Uncertainty Quantification for Tool-Using LLMs

April 2024 - October 2024

- Developed a novel method for quantifying the uncertainty of Large Language Model (LLM) question-answering systems that use external tools such as RAG or querying ML models.
- Motivated from increasing need of safety in deployments of LLMs in critical domains (e.g. medicine).
- Presented results at the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025).

Open-world Agent Modeling

May 2023 - September 2023

- Developed a novel method for online modeling of agent behavior from observations, that accounts for unobservable agent-internal states, crucial in HCI, HRI applications.
- Presented a paper describing the method and its results at the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024).

Accelerating Infeasibility Analysis with Deep Learning

September 2022 - September 2023

- Created a graph neural network-based algorithm for accelerating infeasibility analysis of boolean constraint satisfaction problems.
- Achieved more than two orders of magnitude improvement on MUS enumeration for hard problems and presented findings at AISTATS 2024.

Teaching Experience

Reinforcement Learning

Teaching Assistant, Tufts University

Fall 2023

- Tabular Methods, Policy Gradient, Actor-Critic

Artificial Intelligence

Teaching Assistant, Tufts University

Spring 2025

- Logic & Planning, Bayesian Networks, Neural Networks

Probabilistic Robotics

Teaching Assistant, Tufts University

Spring 2024

- Kalman & Particle Filters, Planning & RL, ROS

Generative Models

Teaching Assistant, Tufts University

Fall 2025

- Probabilistic modeling, Autoregressive models & Transformers, Diffusion

Learning From Limited Labeled Data

Teaching Assistant, Tufts University

Fall 2024

- Transfer, Self/Semi-Supervised, Few-shot Learning

Publications

Lymperopoulos, P., & Liu, L. (2025). Act-to-ground: A framework for symbol grounding in planning domains. 19th International Conference on Neurosymbolic Learning and Reasoning [**NeSy 2025**].

Lymperopoulos, P., & Sarathy, V. (2025). Tools in the Loop: Quantifying Uncertainty of LLM Question Answering Systems That Use Tools. In Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems [**AMAS 2025**] (pp. 2645–2647). International Foundation for Autonomous Agents and Multiagent Systems.

Lymperopoulos, P., Goel, S., Thielstrom, R., Krause, E., Feeney, P., Lorang, P., Schneider, S., Wei, Y., Kildebeck, E., Goss, S., et al. (2024). A neurosymbolic cognitive architecture framework for handling novelties in open worlds. [**Artificial Intelligence**], 104111.

Lymperopoulos, P., & Liu, L. (2024). Graph pruning for enumeration of minimal unsatisfiable subsets. International Conference on Artificial Intelligence and Statistics [**AISTATS 2024**], 2647-2655.

Cynthia Feeney, Sarah Schneider, **Panagiotis Lymperopoulos**, Liping Liu, Matthias Scheutz, & Michael C Hughes (2023). NovelCraft: A Dataset for Novelty Detection and Discovery in Open Worlds. Transactions on Machine Learning Research [**TMLR**].

Panagiotis Lymperopoulos, L. L., Yukun Li. (2022). Exploiting variable correlation with masked modeling for anomaly detection in time series. [**NeurIPS 2022**] Workshop on Robustness in Sequence Modeling. <https://openreview.net/forum?id=TCJuzs585W>

Brummer, A. B., **Lymperopoulos, P.**, Shen, J., Tekin, E., Bentley, L. P., Buzzard, V., Gray, A., Oliveras, I., Enquist, B. J., & Savage, V. M. (2021). Branching principles of animal and plant networks identified by combining extensive data, machine learning and modelling. [**Journal of the Royal Society Interface**], 18(174), 20200624.

Gopalakrishnan, S., Soni, U., Thai, T., **Lymperopoulos, P.**, Scheutz, M., & Kambhampati, S. (2021). Integrating planning, execution and monitoring in the presence of open world novelties: Case study of an open world monopoly solver. [**CoRR**], <https://arxiv.org/abs/2107.04303>

Lee, A. H., **Lymperopoulos, P.**, Cohen, J. T., Wong, J. B., & Hughes, M. C. (2021). Forecasting covid-19 counts at a single hospital: A hierarchical bayesian approach. Poster in [**ICLR 2021**] Workshop on Machine Learning for Preventing and Combating Pandemics, arXiv preprint arXiv:2104.09327.

Polleys, C. M., **Lymperopoulos, P.**, Thieu, H.-T., Genega, E., Liu, L., & Georgakoudi, I. (2021). Deep-learning-based image restoration of depth-resolved, label-free, two-photon images for the quantitative morphological and functional characterization of human cervical tissues. [**Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues**] XIX, 11647, 116470Z.

Lymperopoulos, P., Qiu, H., & Min, B. (2020). Concept wikification for COVID-19. Proceedings of the 1st Workshop on NLP for COVID-19 (Part 2) at [**EMNLP 2020**]. <https://www.aclweb.org/anthology/2020.nlpCOVID19-2.29>